

WaveletFlow: Spatially-Localized Frequency Mixing for Flow Matching in PDE Turbulence Modeling

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Abstract

*Current generative models for turbulent flows underrepresent fine-scale structures (vortices, shear layers) due to spectral bias. The current best approach, FourierFlow [18], boosts high frequencies using Fourier analysis, but this process is globally defined - it cannot tell where fine structures are. We propose **WaveletFlow**, replacing Fourier processing with wavelet-based processing that is both frequency-aware and spatially localized. This process selectively enhances details near vortices without amplifying noise elsewhere. On PDEBench [15] compressible Navier-Stokes, WaveletFlow with the db4 wavelet reduces RMSE from 0.3225 to 0.2955 relative to FourierFlow (8.4% improvement), and the Haar variant achieves 0.2974 (7.8% improvement), with both variants also improving nRMSE over the baseline (0.3924 and 0.3865 vs. 0.4201). These gains arise from a drop-in replacement of a single component while keeping the architecture and training regiment of FourierFlow fixed.*

1. Introduction

Turbulent flows exhibit a rich hierarchy of structures spanning many spatial scales, from large-scale currents and eddies down to tiny vortices and thin shear layers. Faithfully predicting these multi-scale dynamics is critical for applications in aircraft design, weather forecasting, and oceanography [15]. Traditional numerical solvers for the governing Navier-Stokes equations can achieve high accuracy but are prohibitively expensive: direct numerical simulation of turbulence scales as $\mathcal{O}(\text{Re}^3)$ [13], rendering real-time or large-scale studies impractical. This computational bottleneck has motivated a growing body of work on data-driven surrogate models [4, 8, 9, 16] and, more recently, generative approaches [2, 14, 18] that aim to capture the full distribution of turbulent states rather than a single deterministic prediction.

Among generative frameworks, diffusion and flow

matching models [5, 10, 11] have shown particular promise for turbulence due to their capacity to capture complex, multi-scale distributions. However, a fundamental limitation persists: *spectral bias*. As established theoretically and empirically by Wang *et al.* [18], the forward noising process corrupts high-frequency signal components earlier than low-frequency ones due to their smaller signal-to-noise ratios. Consequently, the reverse generative process preferentially recovers coarse, large-scale patterns and underrepresents the fine-scale structures (vortices and shear layers) that carry critical physical information. This bias not only reduces visual fidelity but undermines physical consistency, since turbulence demands strict preservation of energy across all scales.

We propose **WaveletFlow**, which replaces Fourier processing with *spatially-localized frequency mixing* based on the discrete wavelet transform (DWT). Our **Wavelet Mixing (WM)** branch applies a multi-level 2D DWT, transforms detail coefficients through per-scale MLPs with learnable scale-dependent reweighting, and reconstructs via the inverse DWT. Because wavelet coefficients retain spatial position, amplification targets only regions of fine-scale activity. This design is supported by prior wavelet-based PDE methods [3, 6, 7, 17], but none combine wavelets with flow matching for turbulence. WaveletFlow fills this gap.

Our contributions are as follows:

- We identify spatial locality as a missing ingredient in frequency-aware generative turbulence modeling and propose Wavelet Mixing, which provides joint spatial-frequency selectivity through the discrete wavelet transform.
- We design WaveletFlow as a drop-in replacement for FourierFlow’s Fourier Mixing branch, selectively enhancing fine-scale features at the spatial locations where they occur without amplifying noise in quiescent regions.
- We show consistent improvements in both RMSE and nRMSE across wavelet families and provide ablations over wavelet choice and decomposition depth.

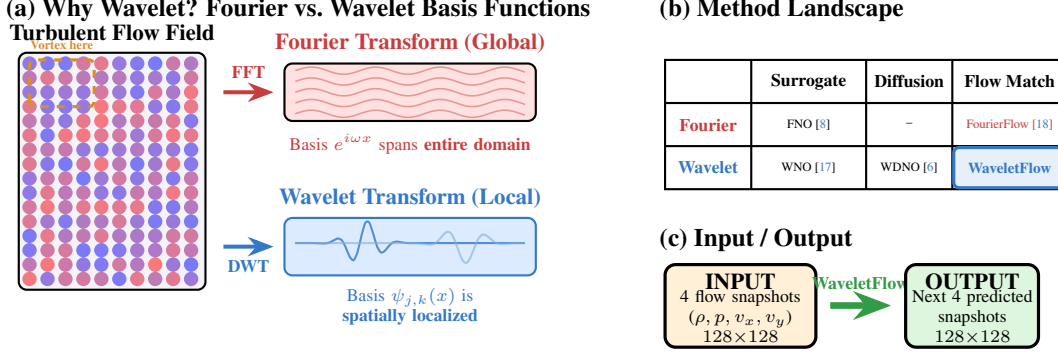


Figure 1. **Motivation and Problem Setting.** (a) Fourier vs. Wavelet transforms. (b) WaveletFlow fills the gap in the method landscape. (c) Input/Output dimensions.

2. Background

2.1. Problem Formulation

We consider turbulent flow governed by a PDE on $[0, T] \times \mathcal{D} \subset \mathbb{R} \times \mathbb{R}^d$:

$$\frac{\partial u}{\partial t} = \mathcal{F}\left(u, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \dots\right) + f(\tau, x), \quad (\tau, x) \in [0, T] \times \mathcal{D}, \quad (1)$$

with initial condition $u(0, x) = u_0(x)$ and boundary condition $\mathcal{B}[u](\tau, x) = 0$ on $\partial\mathcal{D}$, where $u : [0, T] \times \mathcal{D} \rightarrow \mathbb{R}^n$ is the solution field and $f(\tau, x)$ is a forcing term. For compressible Navier-Stokes, u comprises density ρ , pressure p , and velocity components (v_x, v_y) , each discretized on a 128×128 grid, yielding four-channel spatial snapshots.

Following FourierFlow [18], we adopt a *multi-step generative* formulation. Given an observed trajectory of k initial frames $\mathcal{U}_0^k = \{u_0, u_1, \dots, u_{k-1}\}$, the goal is to generate the next k frames $\hat{\mathcal{U}}_k^{2k} = \{\hat{u}_k, \dots, \hat{u}_{2k-1}\}$ directly from Gaussian noise:

$$\hat{\mathcal{U}}_k^{2k} = \mathcal{G}_\theta(\epsilon_{k:2k-1}, \mathcal{U}_0^k), \quad \epsilon_\tau \sim \mathcal{N}(0, \sigma_\tau^2 I), \quad (2)$$

where $\mathcal{G}_\theta$ is a learnable generative model conditioned on the observed past, and $k = 4$ throughout this work.

2.2. Conditional Flow Matching

Our generative backbone is built on conditional flow matching (CFM) [10], which learns a continuous-time velocity field $v_\theta(x, t)$ that deterministically transports samples from a base distribution p_0 (e.g., Gaussian noise) to the data distribution p_{data} via an ordinary differential equation:

$$\frac{dx(t)}{dt} = v_\theta(x(t), t), \quad x(0) \sim p_0, \quad x(1) \sim p_{\text{data}}. \quad (3)$$

The velocity field is trained by regressing onto a target velocity $v^*(x, t)$ derived from linear interpolation between

noise and data samples:

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t \sim \mathcal{U}(0,1), x(t)} [\|v_\theta(x(t), t) - v^*(x(t), t)\|^2], \quad (4)$$

where $x(t) = (1-t)x_0 + tx_1$ with $x_0 \sim p_0$ and $x_1 \sim p_{\text{data}}$. Compared to diffusion models such as DDPM [5], flow matching avoids discrete iterative denoising and provides an efficient, continuous-time transport that is well suited to physics-informed generation [11, 18].

2.3. Spectral Bias in Generative Models

A key theoretical result underpinning our work is the *spectral bias* of generative models, formalized by Wang *et al.* [18]. Consider the forward diffusion process $dx_t = g(t)dw_t$, where w_t is a Wiener process. The signal-to-noise ratio (SNR) at frequency ω evolves as:

$$\text{SNR}(\omega) = \frac{|\hat{x}_0(\omega)|^2}{\int_0^t |g(s)|^2 ds}, \quad (5)$$

where $\hat{x}_0(\omega)$ is the Fourier transform of the initial signal. Since the noise variance (denominator) is constant across frequencies while the signal power of natural and physical signals decays as $|\hat{x}_0(\omega)|^2 \propto |\omega|^{-\alpha}$, higher frequencies reach the noise-dominated regime earlier. The time at which frequency ω is corrupted below a threshold γ satisfies $t_\gamma(\omega) \propto |\omega|^{-\alpha}$ [18]. This implies that the reverse generative process inherently reconstructs low-frequency structures first and struggles to recover fine-scale features such as vortices and shear layers, which reside at high wavenumbers.

2.4. FourierFlow and Its Spatial Limitation

FourierFlow [18] addresses spectral bias through a dual-branch architecture. The *Salient Flow Attention* (SFA) branch employs a differential attention mechanism [19] to suppress common-mode noise and amplify regions of strong relative variation. The *Fourier Mixing* (FM) branch

applies a learnable spectral filter in the Fourier domain:

$$\mathcal{K} \cdot u^l(t, x) = \mathcal{F}^{-1} [W_\theta^l(\xi) \cdot \mathcal{F}[u^l](\xi)], \quad (6)$$

where $W_\theta^l(\xi) = (\beta_\theta^l + \alpha_\theta^l \cdot \|\xi\|^\eta) \cdot \tilde{W}_\theta^l$ assigns larger weights to higher frequencies via $\|\xi\|^\eta$. The two branches are combined through an adaptive gating mechanism, and a pretrained MAE surrogate provides implicit high-frequency guidance via feature alignment.

While effective, the FM branch inherits a fundamental property of the Fourier transform: each basis function $e^{i\xi \cdot x}$ has support over the *entire* spatial domain. The weight $W_\theta^l(\xi)$ modulates a given frequency *uniformly across all spatial locations*. In turbulent flows, however, high-frequency content is concentrated near localized structures, such as vortex cores, shear layers, and shock fronts, while large portions of the domain may be relatively smooth. This mismatch motivates a transform that provides joint spatial-frequency resolution.

2.5. Wavelet Transform

The discrete wavelet transform (DWT) [12] decomposes a signal using basis functions $\psi_{j,k}(x) = 2^{j/2} \psi(2^j x - k)$ that are localized in both frequency (controlled by scale j) and space (controlled by translation k). A J -level 2D DWT decomposes an image into one set of approximation coefficients a_J and $3J$ sets of detail coefficients $\{d_j^{(s)}\}$ for $j = 1, \dots, J$ and $s \in \{\text{LH}, \text{HL}, \text{HH}\}$, capturing horizontal, vertical, and diagonal edges at each scale respectively.

Two properties make wavelets particularly attractive for turbulence modeling. First, **spatial localization**: a large detail coefficient $d_j^{(s)}[k]$ indicates fine-scale activity at a *specific spatial location* k , unlike a Fourier coefficient which reflects global energy at a given frequency. Second, **multi-resolution**: the hierarchical decomposition naturally mirrors the energy cascade in turbulence, where energy transfers from large scales to progressively finer ones.

These properties have been exploited in neural PDE solvers. WNO [17] parameterizes integral operators in the wavelet domain, U-WNO [7] augments this with U-Net skip connections, MWT [3] extends the approach to multi-wavelets, and WDNO [6] applies wavelet-domain processing within diffusion models for PDEs with sharp transitions. Our work is the first to integrate wavelet-based frequency mixing into a flow matching framework for generative turbulence modeling, combining the spatial selectivity of wavelets with the efficiency and physical consistency of continuous-time generative transport.

3. Method

We now describe WaveletFlow, our proposed framework for spatially-localized frequency-aware generative turbulence modeling. WaveletFlow retains the overall dual-branch flow

matching architecture of FourierFlow [18], including the Salient Flow Attention (SFA) branch, cross-attention conditioning, and frequency-aware adaptive fusion, but replaces the global Fourier Mixing branch with a *Wavelet Mixing* branch that operates in a spatially-localized frequency domain.

We first describe the Wavelet Mixing branch in §3.1, the full architecture in §3.2, and then the training objective in §3.3.

3.1. Wavelet Mixing Branch

The Wavelet Mixing branch processes features through four steps: forward DWT, scale-dependent nonlinear processing of the detail coefficients, learnable frequency-aware reweighting, and inverse DWT with a residual connection. The approximation coefficients pass through unchanged.

Multi-level 2D DWT. Given a feature map $x \in \mathbb{R}^{B \times C \times H \times W}$, we apply a J -level 2D DWT (default $J=2$, Daubechies-4 wavelet, zero-padding):

$$(a_J, \{d_j^{(s)}\}_{j=1, \dots, J}^{s \in \{\text{LH}, \text{HL}, \text{HH}\}}) = \text{DWT}_J(x), \quad (7)$$

where $a_J \in \mathbb{R}^{B \times C \times H_J \times W_J}$ holds the coarse approximation coefficients and each $d_j^{(s)} \in \mathbb{R}^{B \times C \times H_j \times W_j}$ holds detail coefficients at scale j for subband s , with $H_j = H/2^j$ and $W_j = W/2^j$. The three subbands capture horizontal (LH), vertical (HL), and diagonal (HH) edges. Unlike Fourier coefficients, each $d_j^{(s)}[k]$ is tied to a specific spatial location k .

Scale-dependent nonlinear processing. At each level j , we concatenate the three subbands along the channel axis to form $\bar{d}_j \in \mathbb{R}^{B \times 3C \times H_j \times W_j}$ and apply a per-level two-layer MLP, implemented as pointwise (1×1) convolutions:

$$\text{MLP}_j(\bar{d}_j) = W_2^{(j)} \sigma(W_1^{(j)} \bar{d}_j), \quad (8)$$

with $W_1^{(j)} \in \mathbb{R}^{3C_{\text{hid}} \times 3C}$, $W_2^{(j)} \in \mathbb{R}^{3C \times 3C_{\text{hid}}}$, hidden width $C_{\text{hid}} = 4C$, and GELU activation σ . Each level has its own parameters, so the MLP learns subband-aware, spatially-varying transformations of the detail coefficients—a capability absent from the global filter used in FourierFlow.

Frequency-aware reweighting. Before reconstruction, each subband is scaled by a learnable factor that biases the branch toward fine scales:

$$\tilde{d}_j^{(s)} = (\beta_j + \alpha_j \cdot 2^{j \cdot \eta_j}) \cdot \text{MLP}_j(\bar{d}_j)|_s, \quad (9)$$

where $\alpha_j, \beta_j, \eta_j$ are scalar parameters per level. The $2^{j \cdot \eta_j}$ term grows exponentially with j , providing a tunable bias

toward finer scales. We initialize $\beta_j = 1$, $\alpha_j = 0$, $\eta_j = 1$, so the weighting factor starts at 1 and the model learns to amplify detail coefficients at scales where high-frequency content is most informative. This plays the role of the $\|\xi\|^\eta$ weighting in FourierFlow’s filter (Eq. 6), but acts on spatially-localized subbands rather than global Fourier modes.

Inverse DWT and residual connection. The branch reconstructs the output from the enhanced details and the unmodified approximation, with a residual link to the input:

$$\text{WM}(x) = \text{IDWT}(a_J, \{\tilde{d}_j^{(s)}\}_{j,s}) + x. \quad (10)$$

The residual stabilizes gradients and lets the branch learn refinements rather than full feature replacements.

Why spatial locality matters. Consider a field with a vortex in one region and smooth flow elsewhere. The Fourier transform spreads the vortex’s energy across all spatial Fourier coefficients at the relevant frequencies, so amplifying those frequencies boosts the vortex and the quiet region alike. The DWT instead concentrates the vortex’s signature in a few detail coefficients localized near it. Equations 8–9 therefore enhance exactly those coefficients and leave quiescent regions undisturbed.

3.2. Overall Architecture

Figure 2 shows the full pipeline, which proceeds in three stages: conditioning, dual-branch feature extraction, and adaptive fusion.

Conditioning. The observed trajectory $\mathcal{U}_0^k$ and the noisy target $x(t)$ are independently tokenized by a shared patch embedding layer (patch size $p \times p$, giving $(H/p) \times (W/p)$ tokens per frame) and augmented with learnable spatiotemporal positional embeddings. Cross-attention (8 heads, head dim 64) then injects the condition tokens into the noisy-target tokens to produce the initial features.

Dual-branch processing. Two parallel branches process the conditioned features.

(i) *Salient Flow Attention (SFA) branch.* This branch stacks L_s spatial transformer blocks followed by L_t temporal transformer blocks, all using the differential attention of FourierFlow [18, 19]. Spatial blocks apply Salient Flow Attention (SF-Attn) independently per frame:

$$\text{SF-Attn}(X) = (\text{Attn}_1(X) - \lambda \tilde{\text{Attn}}_2(X)) V, \quad (11)$$

where Attn_1 captures local structure and $\tilde{\text{Attn}}_2$ models background context using locally mean-subtracted keys. Concretely, the key tensor K_2 is reshaped to the 2D grid

of patch tokens, a depthwise 3×3 average pooling (stride 1, padding 1) computes a local mean $\bar{K}_2$, and the attention then uses $K_2 - \bar{K}_2$ as its keys. Subtracting the local mean suppresses common-mode signals and amplifies regions of strong relative variation - precisely the vortex cores, shear layers, and shock fronts that matter for turbulence. Temporal blocks apply SF-Attn across frames at each spatial location. All blocks use adaptive layer normalization (adaLN-Zero) conditioned on the diffusion timestep embedding.

(ii) *Wavelet Mixing (WM) branch.* This branch stacks L_w WaveletBlock layers. Each block applies a LayerNorm, the WM operator from §3.1, a residual connection, a second LayerNorm, and a pointwise MLP with DropPath regularization:

$$\begin{aligned} x' &= \text{WM}(\text{LN}_1(x)) + x, \\ x'' &= \text{MLP}(\text{LN}_2(x')) + x'. \end{aligned} \quad (12)$$

The branch operates on a detached copy of the conditioned features so it can specialize in frequency-domain processing without affecting the gradients of the SFA branch.

Frequency-aware adaptive fusion. A learned spatial gate combines the two branch outputs u_{SFA} and u_{WM} . Both outputs are projected to the pixel domain by a shared linear head, concatenated along the temporal axis, and passed through a 1×1 convolution with sigmoid activation to produce a per-pixel gating map:

$$G = \sigma(\text{Conv}_{1 \times 1}([u_{\text{SFA}}, u_{\text{WM}}])), \quad (13)$$

$$u_{\text{fused}} = G \odot u_{\text{SFA}} + (1 - G) \odot u_{\text{WM}}, \quad (14)$$

where $[\cdot, \cdot]$ denotes concatenation and $\odot$ is element-wise multiplication. The gate lets the model lean on the WM branch where high-frequency activity is concentrated and on the SFA branch where broader spatial context matters.

3.3. Training Objective

WaveletFlow is trained with a conditional flow matching loss plus a representation-alignment regularizer:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{CFM}} + \gamma \cdot \mathcal{L}_{\text{Align}}. \quad (15)$$

Flow matching loss. $\mathcal{L}_{\text{CFM}}$ (Eq. 4) trains the velocity field v_θ to transport Gaussian noise to the data distribution along a linear stochastic interpolant. The noisy input at time t is $x(t) = (1 - t)x_0 + t\epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, and the target velocity is $v^* = \epsilon - x_0$.

Alignment loss. $\mathcal{L}_{\text{Align}}$ encourages intermediate SFA features (extracted after layer L_e) to match those of a frozen MAE-ViViT encoder [1], a spatiotemporal ViViT ($d=256$, depth 4, 4 heads) pretrained via masked autoencoding on fluid simulation data. A three-layer MLP projector ϕ (hidden dim 2048, output dim 256, SiLU activations) maps the

(d) WaveletFlow Architecture

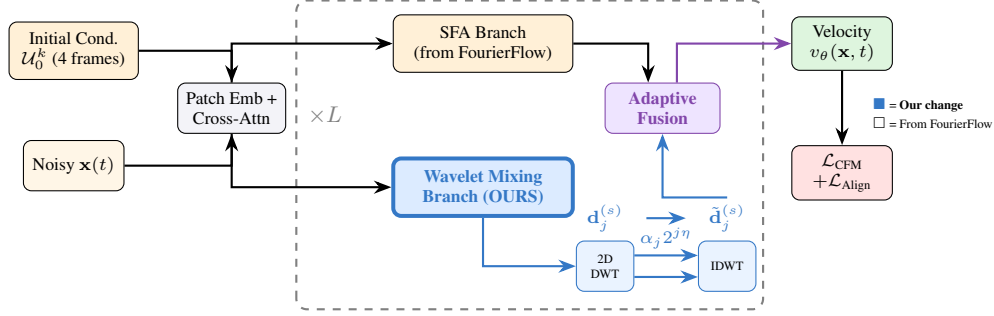


Figure 2. **Architecture details.** We replace the Fourier Mixing branch with our localized Wavelet Mixing branch. The rest of the backbone remains consistent with FourierFlow.

SFA features into the encoder’s embedding space, and the two are compared after ℓ_2 normalization:

$$\mathcal{L}_{\text{Align}} = \mathbb{E} \left[\left\| \frac{\phi(z_{\text{SFA}}^{(L_e)})}{\|\phi(z_{\text{SFA}}^{(L_e)})\|} - \frac{z_{\text{MAE}}}{\|z_{\text{MAE}}\|} \right\|^2 \right], \quad (16)$$

where z_{MAE} is the frozen encoder’s output. As observed in FourierFlow [18], MAE pretraining on fluid data yields representations biased toward high-frequency features, making it a natural target for fine-scale structure recovery. We set $\gamma = 10^{-4}$.

4. Experiments

We evaluate whether replacing global Fourier mixing with spatially-localized wavelet mixing improves generative turbulence modeling. All comparisons are controlled: WaveletFlow uses the same conditional flow matching backbone, conditioning mechanism, SFA branch, adaptive fusion, optimizer, and sampling procedure as FourierFlow [18]. The only changed component is the frequency mixing branch.

4.1. Dataset

We use the 2D compressible Navier–Stokes benchmark from PDEBench [15].

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (17)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \frac{1}{\rho} \nabla \cdot \boldsymbol{\tau}, \quad (18)$$

$$\frac{\partial p}{\partial t} + \mathbf{u} \cdot \nabla p + \gamma p \nabla \cdot \mathbf{u} = (\gamma - 1) [\boldsymbol{\tau} : \nabla \mathbf{u} - \nabla \cdot \mathbf{q}]. \quad (19)$$

Here, ρ is the density, $\mathbf{u} = (u, v)$ is the velocity vector, p is the pressure, γ is the ratio of specific heats, $\boldsymbol{\tau}$ is

the viscous stress tensor, and $\mathbf{q}$ is the heat flux. Each trajectory contains four physical fields: density ρ , pressure p , and velocity components (v_x, v_y) . Following the multi-step setup in §2.1, the model observes $k = 4$ initial frames and generates the next $k = 4$ frames. The original 512×512 simulations are spatially downsampled to 64×64 used by the model due to the limitation of compute resource, and all fields are normalized with statistics computed from the training split.

4.2. Baselines and Variants

We compare three frequency-mixing choices:

- **FourierFlow:** the AFNO/Fourier mixing baseline from Wang *et al.* [18].
- **WaveletFlow-db4:** our Wavelet Mixing branch with a Daubechies-4 wavelet basis.
- **WaveletFlow-Haar:** our Wavelet Mixing branch with a Haar wavelet basis.

Both WaveletFlow variants use a two-level DWT ($J = 2$) and the same per-level detail-coefficient MLP described in Eq. 8. Thus, the db4 and Haar comparison isolates the effect of the wavelet basis while holding the architecture fixed. All the models were trained for 1600 epochs.

4.3. Metrics

We report root mean squared error (RMSE) and normalized RMSE (nRMSE) over the generated future frames. RMSE measures absolute prediction error in normalized physical fields, while nRMSE divides the error by the target magnitude and is therefore more sensitive to relative errors in low-energy regions. Lower values are better for both metrics. We also report the relative RMSE improvement over FourierFlow:

$$\Delta_{\text{RMSE}} = \frac{\text{RMSE}_{\text{FourierFlow}} - \text{RMSE}_{\text{Method}}}{\text{RMSE}_{\text{FourierFlow}}} \times 100\%. \quad (20)$$

Table 1. Main comparison on PDEBench compressible Navier–Stokes. Lower is better for RMSE and nRMSE. Relative improvement is computed against FourierFlow.

Method	RMSE ↓	nRMSE ↓	RMSE Improv. ↑
FourierFlow	0.3225	0.4201	—
WaveletFlow-db4	0.2955	0.3924	8.4%
WaveletFlow-Haar	0.2974	0.3865	7.8%

4.4. Implementation Details

All models are trained with the conditional flow matching objective in Eq. 4 plus the MAE alignment regularizer in Eq. 15. We use the same number of observed and generated frames ($k = 4$), the same patch embedding, and the same adaptive fusion module across all methods. WaveletFlow uses zero-padded DWT/IDWT operations and processes only the high-pass detail coefficients; the low-pass approximation coefficients are passed to the inverse transform unchanged. During evaluation, samples are generated from Gaussian noise with an Euler ODE sampler. Unless otherwise stated, the number of sampling steps is fixed across methods. The MAE pretraining models were trained with NVIDIA GeForce RTX 3090 GPU, and WaveletFlow and FourierFlow training were conducted on $3 \times$ NVIDIA RTX6000 Ada GPUs.

4.5. Main Results

Table 1 shows that both wavelet variants improve RMSE and nRMSE over FourierFlow. WaveletFlow-db4 obtains the best RMSE, reducing error from 0.3225 to 0.2955, an 8.4% relative improvement, while also lowering nRMSE from 0.4201 to 0.3924. WaveletFlow-Haar achieves a comparable RMSE of 0.2974 (7.8% improvement) with the best nRMSE of 0.3865. These results suggest that wavelet-based mixing improves both absolute reconstruction of localized structures and relative accuracy across the field.

WaveletFlow-db4 achieves the best RMSE at 0.2955 (8.4% improvement), while also reducing nRMSE to 0.3924 from FourierFlow’s 0.4201. WaveletFlow-Haar achieves a comparable RMSE of 0.2974 with the best nRMSE of 0.3865. The difference between the two variants is consistent with their spatial-frequency properties: db4 has smoother, longer-support filters that provide better frequency selectivity, while Haar’s compact support is well matched to sharp, spatially concentrated features such as shocks and thin shear layers. Moreover, Figures 2 and 3 clearly show that WaveletFlow predicts vortical structures, especially the extrema in the velocity field, more accurately than the standard FourierFlow. This improved performance suggests that WaveletFlow is better able to capture high-frequency fluid fluctuations.

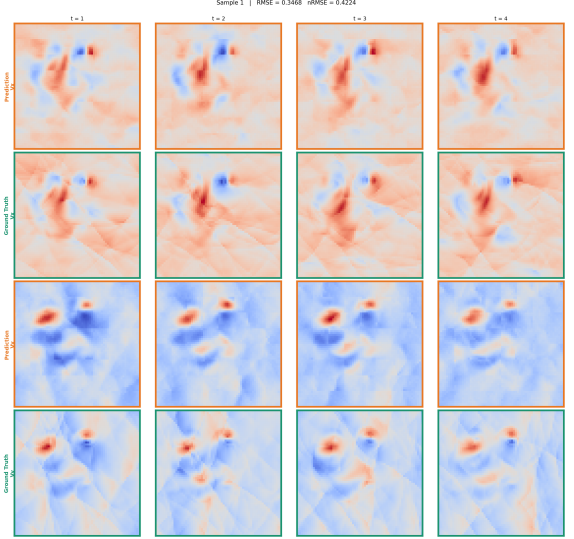


Figure 3. Results of flow field based on FourierFlow algorithm

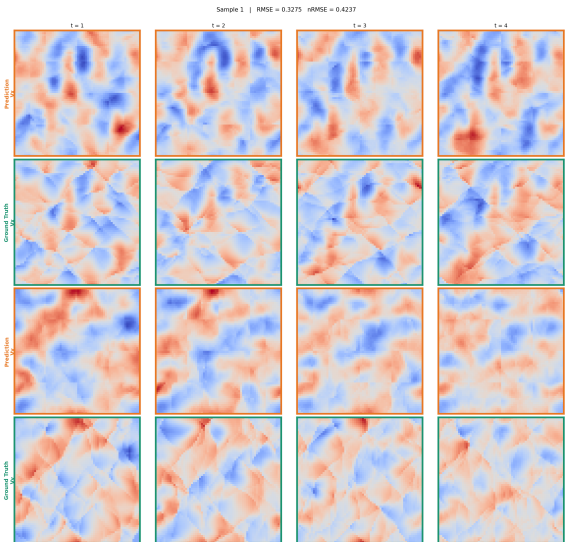


Figure 4. Results of flow field based on WaveletFlow (Harr) algorithm

4.6. Wavelet Family Ablation

Besides all the models mentioned above, we trained another hybrid architecture, HybridFlow, that combines both FourierFlow and WaveletFlow, training and testing with the same procedure. HybridFlow extends FourierFlow by incorporating a wavelet-based residual refinement module after the AFNO (Adaptive Fourier Neural Operator) spatial mixing stage. Specifically, given the feature map produced by the AFNO blocks, a parallel stack of WNO2D (Wavelet Neural Operator) blocks is applied and its output is added back as a residual correction:

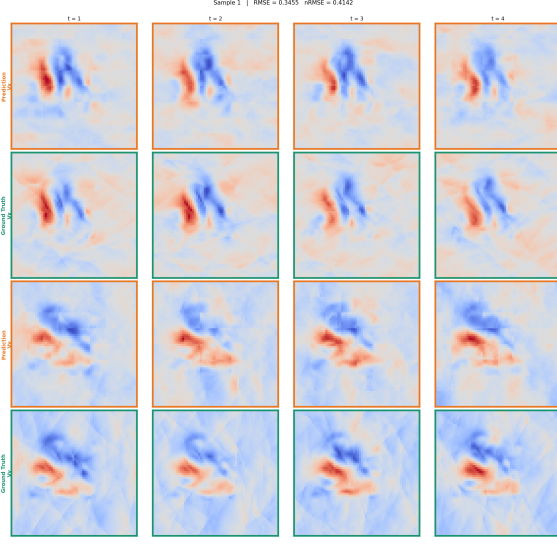


Figure 5. Results of flow field based on HybridFlow algorithm

$$\mathbf{f}_{\text{hybrid}} = \mathbf{f}_{\text{AFNO}} + \text{WNO}(\mathbf{f}_{\text{AFNO}})$$

By stacking WNO as a residual correction on top of AFNO, HybridFlow allows the model to first capture large-scale flow structure and then refine local, high-frequency details that Fourier mixing alone may miss. Both the AFNO and WNO branches share the same hidden dimension and depth configuration, and the combined representation is passed to the temporal attention blocks and final prediction head unchanged. Due to time constraints, we trained HybridFlow for only 1000 epochs. Nevertheless, it achieved an RMSE of 0.3455 and an nRMSE of 0.4142, which are already better than those of FourierFlow trained for 1600 epochs.

4.7. Qualitative Interpretation

FourierFlow applies frequency amplification globally: increasing a high-frequency mode affects the entire domain. This is helpful when the error is distributed broadly, but it can over-amplify smooth regions when high-frequency activity is concentrated only near vortices, shocks, or shear layers. WaveletFlow instead modifies detail coefficients at particular scales and locations. As a result, it can place more modeling capacity near physically salient fine-scale structures while leaving smooth regions closer to the low-frequency reconstruction. The stronger performance of WaveletFlow-Haar indicates that, for this compressible Navier–Stokes setting, compact spatial support is especially valuable.

5. Conclusion

We presented WaveletFlow, a generative turbulence modeling framework that introduces spatially-localized frequency mixing into flow matching by replacing FourierFlow’s global Fourier Mixing branch with a wavelet-based alternative. Our central hypothesis was that the spatial blindness of Fourier-domain processing leaves performance on the table: turbulent fine-scale structures are localized, and the transform used to enhance them should be too.

We emphasize that WaveletFlow is designed as a *lightweight proof of concept* to validate the spatial-locality hypothesis. The original FourierFlow was trained on $8 \times \text{H800}$ GPUs over an extended period; faithfully replicating that scale was beyond our computational budget. Instead, we adopted a smaller training configuration with fewer epochs and reduced batch size, applying it consistently to both our method and the FourierFlow baseline to ensure a fair comparison under identical resource constraints. Even in this lightweight setting, WaveletFlow with the db4 wavelet reduces full test RMSE from 0.3225 to 0.2955 (8.4% improvement) and the Haar variant achieves 0.2974 RMSE (7.8% improvement), with both variants also improving nRMSE (0.3924 and 0.3865 vs. 0.4201).

Limitations and future work. Several directions remain open. First, given additional compute, training WaveletFlow at the full scale of the original FourierFlow experiments would clarify the ceiling of the locality advantage. Second, we have only explored separable, decimated wavelet transforms; lifting-based or learnable wavelet families [3] could offer additional flexibility. Third, the current WM branch processes a detached copy of the features, preventing gradient flow from the fusion loss back through the wavelet path; relaxing this design choice may unlock further gains. Fourth, scaling to higher-resolution grids and more turbulent regimes (higher Reynolds numbers) may amplify the locality advantage, as finer structures become more spatially concentrated. Finally, extending WaveletFlow to three-dimensional and incompressible flow settings would broaden its practical applicability to domains such as aerodynamic design and climate modeling.

More broadly, our work highlights a simple but under-explored principle for physics-informed generative models: when the structures of interest are spatially localized, the spectral tools used to model them should be too. Even as a proof of concept under constrained resources, WaveletFlow demonstrates that this principle translates into measurable gains, motivating further exploration of spatially-adaptive representations in scientific generative modeling.

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